

DEMO PROJECT

Fast, safe, personalized recommendations

Combining a decision model (Jev) with LLMs: fewer tokens, catalogs that scale, no constraint violations

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THE PROBLEM

Not every place suits every traveler



Elderly or wheelchair user

Basilica Cistern: a long staircase at the entrance



Family with small kids

Kadife Street: bars open late into the night



Pet owner

Topkapı Palace: no pets inside the museum

Any recommendation platform has to filter a large catalog for each traveler: fast, cheap and reliably.

THE SCALE PROBLEM

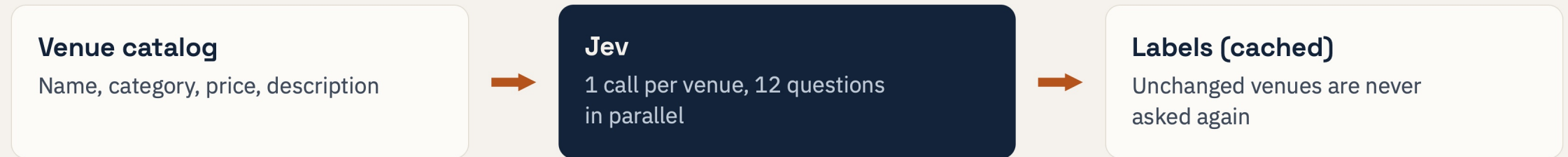
21x

A catalog of 300,000 venues $\approx$ 20.7 million tokens. The largest context windows hold about 1 million.

Sending the whole catalog to an LLM is not just expensive, it does not fit. A pre-filter is required.

The solution: a two-stage design

ONCE, OFFLINE



ON EVERY REQUEST



How I used Jev

A decision model that does not write text: send a state and questions, get back probabilities and confidence.

12 questions in one call

Answered in parallel: 100 calls instead of 1,200.

Constraints vs interests

Constraints exclude ("is there a barrier?"), interests rank ("would they enjoy it?").

Three-way accessibility

Barrier, no barrier, no information. No information means a warning, not a hidden venue.

Numbers stay in code

Structured fields like price are filtered by code, not by Jev.

What the user sees

Jev + LLM Recommendations

Sample venue catalog (Istanbul) · 100 venues pre-labeled with Jev · top 10 sent to the LLM

Route Comparison (full list vs Jev-filtered)

Profile: Elderly traveler + Culture enthusiast

Summary

Total venues	Excluded	Flagged (among remaining)	Sent to the model
100	33	45	10

Recommended route

#	Venue	District	Interest score	Constraint status
1	Sultanahmet Camii (Mavi Cami)	Sultanahmet	0.96	Elderly traveler: suitable
2	İstanbul Arkeoloji Müzeleri	Sultanahmet	0.96	Elderly traveler: not verified
3	Pera Müzesi	Beyoğlu	0.93	Elderly traveler: suitable

33 excluded

Break a constraint, never sent to the LLM

45 flagged

Shown as "accessibility not verified"

10 sent

Top interest scores; the LLM picks the route

Why was a venue excluded?

Excluded venues (33)

Note: Jev does not generate explanations; it returns only a decision and a probability. The highlights below are keyword matches marked by code and may not be the exact reason for Jev's decision.

▼ Elderly traveler — 33 venues

Ayasofya · historic site · Jev choice: **barrier_present** (confidence 0.61)

Cues in the description:

1,500-year-old domed landmark with Byzantine and Ottoman heritage. Tourists visit the upper gallery via a separate entrance; the ground floor is open for worship. Queues can be long.

Topkapı Sarayı · museum · Jev choice: **barrier_present** (confidence 1.00)

Cues in the description:

Palace of the Ottoman sultans. Large courtyards, the Harem, the treasury and terraces with Bosphorus views. A full visit takes 3-4 hours, with a lot of walking and some **stairs**.

Yerebatan Sarnıcı · historic site · Jev choice: **barrier_present** (confidence 1.00)

Cues in the description:

Underground Byzantine cistern; dim lighting, walkways above the water and the Medusa heads. A long **staircase** leads down at the entrance; inside it is cool and damp.

Decision + confidence

Every exclusion shows Jev's choice, its confidence and the source text.

Meaning, not keywords

Ayasofya has no "stairs" keyword, yet Jev read "upper gallery" as a barrier (confidence 0.61).

Labeling: fast and almost free

\$0.0073

to label all 100 venues

0.43 s

average time per call

100

calls; 1,200 without
parallel questions

1,737

input tokens per call

At 300,000 venues: about \$22, once. After that, only new or changed venues are labeled again.

Accuracy: 85% on 80 decisions

Constraint	Correct	Safe errors	Unsafe errors
Elderly traveler	17/20 (85%)	3	0
Wheelchair user	17/20 (85%)	2	1
Small kids	14/20 (70%)	6	0
Pet owner	20/20 (100%)	0	0
Total	68/80 (85%)	11	1

1 / 80

unsafe error, and it was still shown with a warning

Safe error: an unnecessary warning or exclusion. Unsafe error: a venue with a barrier shown as more suitable than it is. Reference labels for 20 venues were created by an LLM from the original Turkish descriptions, without seeing Jev's output; the catalog was then translated, so the comparison is approximate.

VENDOR COMPARISON

Works with any LLM: two vendors, same result

-88%

input tokens per request, for both vendors

1 → 0

constraint violations per vendor, full list → filtered

-77%

monthly cost with Claude Opus 5.5: about \$7,400 less

Without a catalog, both models broke constraints 4 times and suggested some places that are not in the catalog.

Same prompts, two vendors

Model	Prompt	Input tokens	Time	Violations	Relevance	Monthly cost
GPT-5.6 Luna	A · full list	7,011	7.7 s	1	0.79	\$452
	B · Jev-filtered	842	4.9 s	0	0.88	\$80
	C · no list	94	8.0 s	4	0.81	\$61
Claude Opus 5.5	A · full list	7,011	9.8 s	1	0.82	\$9,583
	B · Jev-filtered	842	8.4 s	0	0.86	\$2,188
	C · no list	94	11.0 s	4	0.88	\$1,502

Averages over 3 profiles, one run each; times measured by hand in the consumer apps. Monthly cost: 10,000 requests/day, list prices, input + estimated output, no prompt caching. C: the model names venues itself; 3 of 11 (Luna) and 1 of 11 (Opus) suggestions were not in the catalog. Violations and relevance are scored with Jev's labels.

The filter makes cheap models usable

8 → 0

constraint violations, 5 profiles

77%

of full-list picks came from the first 10 positions
(10% by chance)

Wheelchair user + food lover

FULL LIST

Archaeology Museums, Museum of Innocence, Dolmabahçe Palace, Kariye Mosque: no food at all

JEV-FILTERED

Kadıköy Fish Market street, Çiya Sofrası and other food venues

A BONUS FINDING

A data-gap report for free

44%

of venues: no accessibility info for elderly travelers

51%

no info for wheelchair users

Flagged venues — accessibility not verified (45)

Not excluded, but the description has no accessibility information (or Jev's confidence is below 0.5). Shown to the user with a warning.

Venue	Category	Interest score	In candidate list
İstanbul Arkeoloji Müzeleri	museum	0.96	✓
Beylerbeyi Sarayı	museum	0.94	✓
Asitane	restaurant	0.94	✓
Kılıç Ali Paşa Hamamı	hammam	0.9	✓
HodjaPasha Kültür Merkezi	show	0.9	
Feriye Lokantası	restaurant	0.89	
Ortaköy Meydanı ve Camii	neighborhood walk	0.88	
Arasta Çarşısı	bazaar	0.85	
Neolokal	restaurant	0.85	
Deraliye	restaurant	0.85	

Not hidden: shown with a warning. The same list tells the catalog team what to complete.

Not only for travel

Any large catalog plus user constraints: Jev pre-filters, the LLM reasons.

Restaurants and food delivery

Dietary needs, kids, step-free entrance

E-commerce

Compatibility, safety for children, shipping limits

Real estate

Pets allowed, elevator, quiet street

Events and tickets

Age limits, accessibility, late hours

Limitations

Sample data

Real venues, but descriptions and prices are illustrative.

Small test sets

20 reference venues; 3 profiles, one run each per vendor.

Scored by Jev

Violations and relevance use Jev's labels; a blind human rating is needed.

Manual timing

Measured by hand in consumer apps, default settings.

Approximate cost

One tokenizer for all, estimated output, no caching.

No geography

Routes can combine districts that are far apart.

Next steps

1. Pilot on a real catalog in one city
2. Blind human rating of routes, full list vs filtered
3. More vendors and repeated runs, with timing measured through the API
4. Three-way question for pets; tune thresholds on a larger test set
5. Group routes by district, with distances computed in code
6. Expose the filter as an MCP tool that any agent can call

SUMMARY

Jev makes the fast, repeated decisions. The LLM does the reasoning.

88% fewer tokens, catalogs that scale, and constraint-safe recommendations with any LLM.

Thank you · Questions?